

Taylor & Francis
Taylor & Francis Group

Australian Journal of Earth Sciences

An International Geoscience Journal of the Geological Society of Australia

ISSN: 0812-0099 (Print) 1440-0952 (Online) Journal homepage: www.tandfonline.com/journals/taje20

Syn-deformational and drilling-induced remanent magnetizations from diamond drill cores of the Mt Eclipse Sandstone, Central Australia

Z. X. Li, C. McA. Powell & D. G. Morris

To cite this article: Z. X. Li, C. McA. Powell & D. G. Morris (1991) Syn-deformational and drilling-induced remanent magnetizations from diamond drill cores of the Mt Eclipse Sandstone, Central Australia, Australian Journal of Earth Sciences, 38:4, 473-484, DOI: 10.1080/08120099108727984

To link to this article: https://doi.org/10.1080/08120099108727984

Published online: 09 May 2007.

Submit your article to this journal

Article views: 41

View related articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=taje20

Australian Journal of Earth Sciences (1991) **38**, 473–484

Syn-deformational and drilling-induced remanent magnetizations from diamond drill cores of the Mt Eclipse Sandstone, central Australia

Z. X. LI[1,*], C. McA. POWELL[1,*] AND D. G. MORRIS[2,†]

[1]*Australian Plate Research Group, School of Earth Sciences, Macquarie University, NSW 2109, Australia.*
[2]*Northern Territory Geological Survey, Alice Springs, NT 0870, Australia.*

Six diamond drill cores of the latest Devonian–Early Carboniferous Mt Eclipse Sandstone from the Ngalia Basin, central Australia, were oriented using regional bedding strike, core orientation, core/bedding angle and magnetic fabric information. Combined alternating field and thermal demagnetization revealed a predominant drilling-induced remanence (C_1), a syn-deformational overprint (C_2) and a possible primary remanence (C_3). The drilling-induced remanence is most likely a combination of isothermal remanent magnetization and piezo-remanent magnetization, formed in an axial symmetrically-distributed strong magnetic field inside the magnetic drill barrel, during the initial drilling process. C_1 has almost totally reset the remanences in magnetite and goethite, but has had little impact on hematite. C_2 was found in four cores and is a chemical overprint formed during the Alice Springs Orogeny. This chemical event could also have been responsible for the uranium mineralization in the rocks. C_2 gives a mid- to Late Carboniferous pole at (29.2°S, 148.1°E) with DP = 45.8°, DM = 47.1°. C_3 was found in five samples in one core.

Key words: Alice Springs Orogeny, drilling-induced remanence, Mt Eclipse Sandstone, Ngalia Basin, overprint, palaeomagnetism.

INTRODUCTION

Palaeomagnetism of the latest Devonian–Early Carboniferous Mt Eclipse Sandstone, Ngalia Basin (Fig. 1), has been previously investigated by Li *et al.* (1989) on surface samples. The original aim of that study was to obtain an Early to mid-Carboniferous palaeomagnetic pole for cratonic Australia. Such a pole would help to refine the poorly defined Carboniferous part of the apparent polar wander path for Gondwana (Schmidt & Embleton 1987; Li *et al.* 1990) and, therefore, to understand better the rapid drift of Gondwana during that period as suggested by palaeoclimatologists (Koppen & Wegener 1924; Caputo & Crowell 1985). Two remanence components were reported from that study: a recent weathering-produced soft (easily demagnetized) component and a steep-downward directed stable component acquired during the mid-Carboniferous deformation. However, no primary magnetic remanence was found. Li *et al.* (1989) attributed this to the lack of detrital magnetic minerals in the rocks, the effect of deformation in the region studied and possible chemical alterations during both the mid-Carboniferous deformation and Recent exposure.

In Li *et al.*'s (1989) study, samples were collected from outcrops in the northern part of the basin, where thrusting and folding produced medium to high limb dips during the mid- to Late Carboniferous. According to Wells and Moss (1983), the intensity of deformation decreases from north to the south in the Ngalia Basin. Thus, one would expect that if there were some pre-deformational magnetic remanence it would have survived in the less-deformed rocks in the southern part of the basin. However, as outcrops are sparse in the southern Ngalia Basin, it was decided to study diamond drill cores recovered by exploration companies in the region. An advantage with diamond drill core materials is that they are commonly fresher than the surface samples, but a drawback is that their original orientation is hard to determine.

SAMPLING AND ORIENTING TECHNIQUES

The Ngalia Basin is an intracratonic basin overlying the Arunta Block, central Australia (Fig. 1a). The Mt Eclipse Sandstone, whose age ranges from latest

*Present address: Geology Department, University of Western Australia, Nedlands, WA 6009, Australia.
†Present address: Pacific Oil & Gas Pty Limited, 10 Wilkinson St, Alice Springs, NT 0870, Australia.

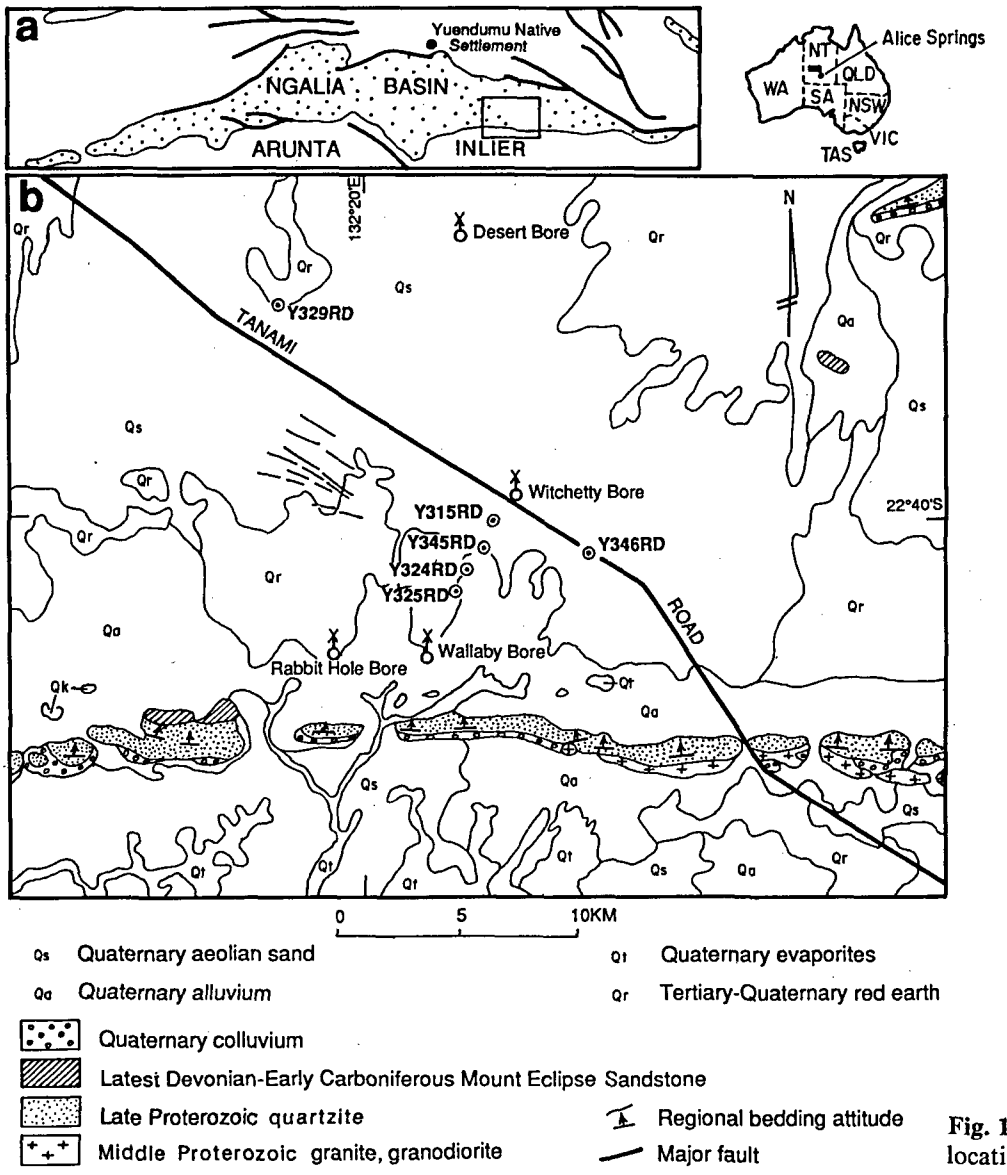


Fig. 1 Regional geology and locations of the drill holes.

Devonian to the Early Carboniferous (see discussion in Li *et al.* 1989), unconformably overlies most of the older rock formations in the basin. It was folded and thrust during the mid- to Late Carboniferous Mt Eclipse phase of the Alice Springs Orogeny (Wells & Moss 1983).

Sampling was carried out in the Core Library of the Northern Territory Geological Survey in Alice Springs, central Australia. Six 36 mm diameter diamond drill cores (hole size 140 mm), all drilled by Agip Australia Pty Ltd from the eastern end of the basin (Fig. 1b), were sampled (Table 1). The core material varies from reduced grey sandstone and siltstone to more oxidized red conglomerate, sandstone and some siltstone and mudstone. Sampling was concentrated on the sandstone, siltstone and minor mudstone interbeds. A simplified log was recorded for these drill cores. The age of the rocks sampled is estimated to be Tournaisian or early Visean on the basis of spores recovered from the drill core (Agip Australia Pty Ltd unpubl. data).

A total of 96 core samples was collected, each 10–30 cm long. For the purpose of orienting the samples, the stratigraphic order and bedding surfaces of the core segments were examined and marked before they were removed from the core shed. Where possible, the stratigraphic facing was determined using small-scale cross-laminae to ensure that the core segments were correctly oriented with respect to the vertical. In some lithologies, particularly those that are coarse grained, independent evidence for facing was not always apparent and the orientation as indicated on the core tray was used. Bedding surfaces were defined using lithological

Table 1 Information about the studied drill cores.

Core name	Location °S	Location °E	Bottom depth (m)	Core orientation (dip/dip-azm)	Estimated bedding (dip/dip-azm)	No. samples
Y315RD	22.67	132.39	540	70/180	104/192	22
Y324RD	22.69	132.38	261	Vertical	14/358	13
Y325RD	22.70	132.37	300	Vertical	10/358	15
Y329RD	22.58	132.31	295	Vertical	32/000	21
Y345RD	22.67	132.39	282	70/000	69/188	17
Y346RD	22.68	132.43	270	Vertical	7/195	8

boundaries and cross-laminae structures. Bedding to core-axis angles was measured and recorded using a simple cylindrical protractor device.

In the laboratory, two specimens of 32 mm height were sliced directly from each core segment (these specimens are referred to as the 'large specimens' below). Whenever the remnants permitted, they were re-drilled along the centre of the core segments to the standard palaeomagnetic specimen diameter of 25 mm and sliced to the standard height of 22 mm.

The bedding strikes were determined through correlation between closely spaced drill holes (B. G. Jones pers. comm. 1987; D. G. Morris unpubl. data). Their dips were determined mainly using the measured core/bedding angles and core orientation information, and checked against the dips obtained through drill hole correlation (B. G. Jones pers. comm. 1987; Table 1).

Of the six drill cores sampled, four were drilled vertically, and two were inclined. To restore samples from the vertical drill cores to their *in situ* orientations simply required rotating the sample around the vertical axis of the drill cores so that the strike of the bedding, as determined from each core sample, was aligned with that of the regional bedding. The true dip of the bedding is simply the complement of the core axis/bedding angle.

For the inclined drill cores the restoring process is somewhat more complicated. The true dip is no longer simply the complement of core axis/bedding angle, and the strike, as determined from each core sample, is an apparent strike, unless the drill core azimuth is perpendicular to the bedding strike. In general, orientation still requires that the bedding strike be aligned with the regional bedding strike. This was achieved by performing two successive rotations: first, restore the drill core to its original azimuth and inclination; then rotate the core around its axis so that the bedding plane on the core sample shares the same dip-azimuth as the regional bedding. Using this method, the true dip of the bedding was obtained.

ANISOTROPY OF MAGNETIC SUSCEPTIBILITY

Laboratory work was carried out in the Rock Magnetism Laboratory of the CSIRO Division of Exploration Geoscience. Anisotropy of low-field magnetic susceptibility (AMS) of 79 samples from the six drill cores was measured using a modified DIGICO anisotropy delineator (Coward 1989). The purpose of these measurements is two-fold. First, if the AMS minimum axes (axes of minimum susceptibility, K_{min}) are parallel to the bedding pole (or poles), it would lend confidence to the identified bedding planes and, therefore, to the orientation method. Second, if the K_{min} axes differ from the bedding pole beyond what might be expected due to reasonable orientation uncertainties, this might be attributable to undetected internal ductile deformation. Such microscopic deformation can manifest itself as a systematic deflection of the K_{min} axes away from the pole of the bedding. The magnetic susceptibility is also useful in identifying the magnetic minerals in samples of different rock type and different colour. All such information is important in the overall data interpretation.

Figure 2 shows stereonet plots of the AMS principal axes. Magnetic foliations are well developed in all the six cores, with the distributions of their K_{min} axes roughly Fisherian (normal spherical). Although a magnetic lineation can be seen clearly in core Y345RD (Fig. 2e), its foliation is still well preserved. The most important feature shown in Fig. 2 is that the mean K_{min} axes are well grouped in all the cores, and none of mean directions is significantly different from the poles to the estimated bedding. Two conclusions that can be drawn from these AMS results are: (1) the sample orientation method is reasonably accurate; (2) no significant internal ductile deformation has occurred in the rock strata sampled, despite some small shears being observed in the drill cores.

By comparing the bulk susceptibility of samples with a different colour, it was found that the values

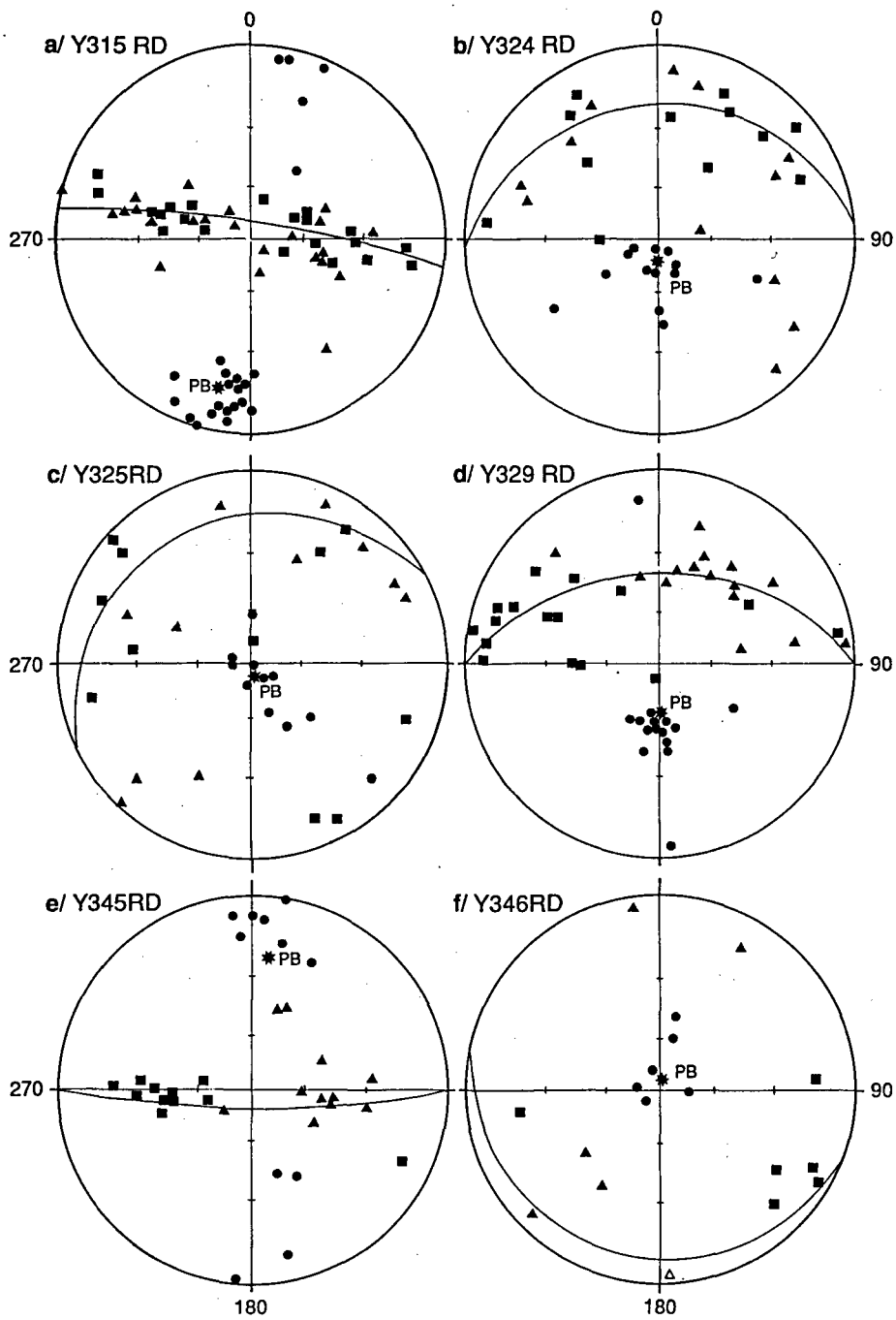


Fig. 2 Lower hemisphere equal-angle projections of the anisotropy of magnetic susceptibility (AMS) data from the six drill cores. Dots represent directions of the minimum axes (K_{min}), squares maximum axes (K_{max}), and triangles intermediate axes (K_{int}). Great circles are the average magnetic foliation planes. PB is the pole to the estimated bedding plane.

generally higher than in the reddish specimens (oxidized). This is probably due to the existence of more magnetite in the grey coloured samples. This suggestion is supported by the higher value of the natural remanent magnetization and the lower unblocking temperatures in the reduced specimens.

DEMAGNETIZATION AND REMANENCE INTERPRETATION

A total of 244 specimens was subjected to detailed demagnetization (151 large specimens and 93 of standard size). All the large specimens, and 23 of the standard specimens, were thermally demagnetized step-wise in up to 15 increments. The furnace used has automatic control and was installed in a magnetic-free space ($\sim \pm 5$ nT) (Coward *et al.* 1985). The remainder of the standard specimens were demagnetized using a Schonstedt GSD1 alternating field demagnetizer to up to 75 mT (750 Oe), and then followed by up to 11 steps of thermal demagnetization. Three spare specimens were subjected to chemical etching. Measurements were made using a three-axis CTF cryogenic magnetometer. Data analysis was carried out by applying the principal-

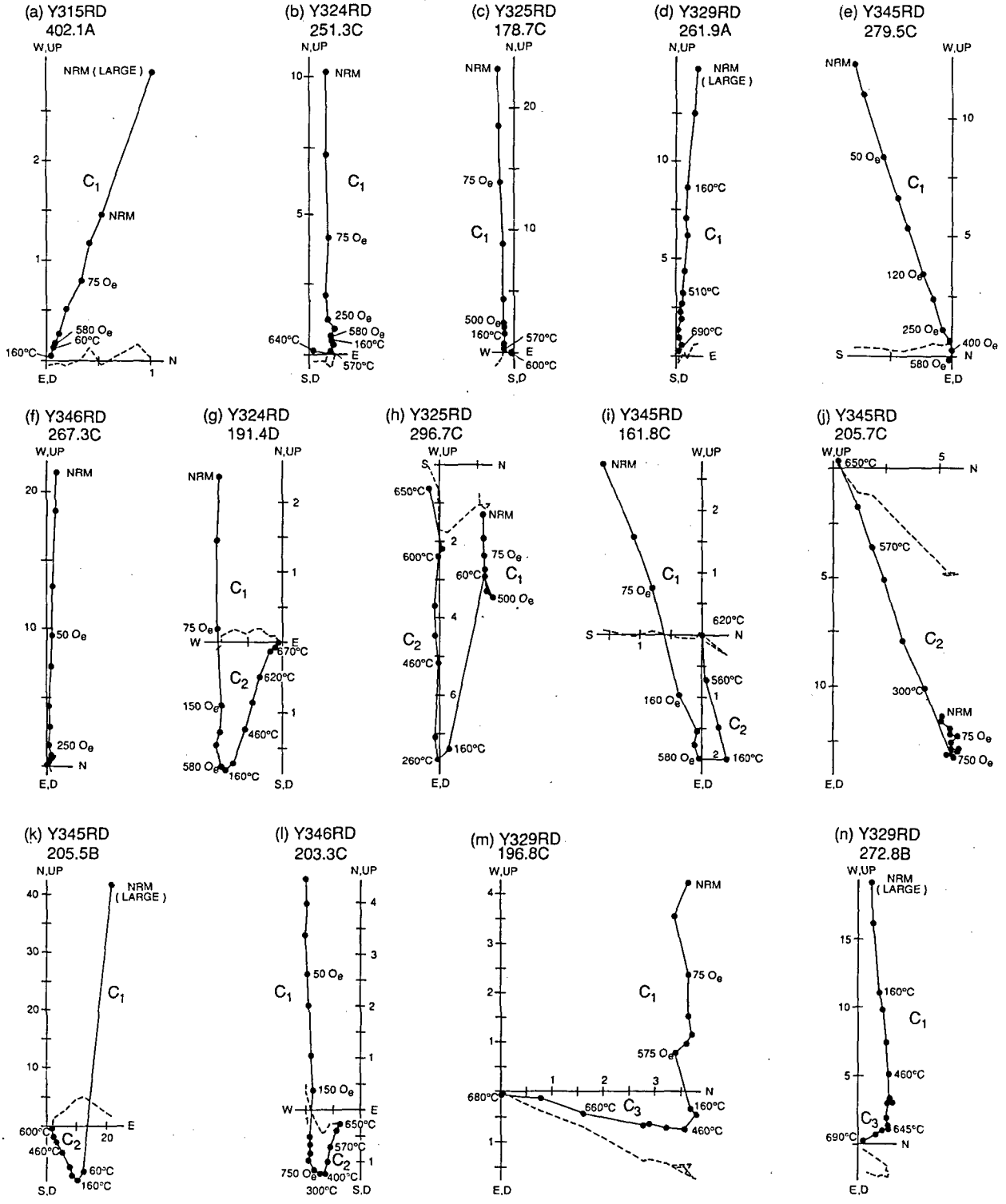


Fig. 3 Representative orthogonal plots of demagnetization data in geographic co-ordinates. Dashed lines are projections on horizontal planes, and solid lines on vertical planes. Each dot represents a demagnetization step. N, E, S and W: north, east, south and west respectively. UP and D: up and down respectively. NRM: natural remanent magnetization. C_1, C_2 and C_3 are the three identified remanent components. All units are in mAm ($\times 10^{-6}$ emu/cm^3).

component analysis technique (Kirschvink 1980) in conjunction with orthogonal plots (Zijderveld 1967).

Three distinct magnetic remanence components were revealed. The first component (C_1) was well developed in all the samples and is generally parallel to the drill core. The second one (C_2) was found in samples from four cores and was directed steep-downwards in geographic co-ordinates. The third component (C_3) was present only in five samples from one drill core, and was directed moderately-upwards in both geographic and stratigraphic co-ordinates. C_2 and C_3 were not found to coexist in any of the specimens, but both were found to coexist with C_1. In specimens having C_1 plus C_2 or C_1 plus C_3, the unblocking temperatures (or coercive forces) of C_2 and C_3 are higher than that of C_1. Typical orthogonal plots of the demagnetization data are presented in Fig. 3. A summary of the remanence directions and palaeomagnetic poles is given in Table 2. More detailed discussion and interpretation of these three components follows.

Drilling-induced remanence

The C_1 component is so dominant in the reduced sandstone and siltstone samples that only 7% (three samples) possessed another component. In the reddish rock samples (sandstone, siltstone and mudstone), C_1 is not so pervasive, but is still a well-defined component in every sample. In the reduced specimens, the unblocking temperatures of C_1 are usually < 580°C, but in the reddish specimens they can extend up to 690°C (Fig. 3d). In both types of rocks C_1 can generally be eliminated by alternating

Table 2 Summary of the magnetic remanence directions from the drill core material.

(a) Drilling-induced remanence (C_1)

		In situ (geographic) co-ordinates				In drill core co-ordinates			
Core	*n*(N)	Dec. (°)	Inc. (°)	k	α_{95} (°)	Dec. (°)	Inc. (°)	k	α_{95} (°)
Y315RD	22	355.0	−73.1	116.8	2.9	204.7	−86.5	116.8	2.9
Y324RD	13	216.0	−87.2	252.5	2.6	216.0	−87.2	252.5	2.6
Y325RD	15	206.9	−88.5	815.9	1.3	206.9	−88.5	815.9	1.3
Y329RD	21	174.9	−84.6	275.5	1.9	174.9	−84.6	275.5	1.9
Y345RD	17	161.5	−76.7	117.4	3.3	029.7	−81.5	117.4	3.3
Y346RD	8	345.8	−88.9	335.7	3.0	345.8	−88.9	335.7	3.0
Mean	(6)	170.8	−89.3	66.0	8.3	170.6	−89.4	268.2	4.1

(b) Syn-deformational overprint (C_2)

		In situ		20% untilting		100% untilting		Common	
Core	*n* (N)	Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	k	α_{95} (°)
Y324RD	7	302.9	87.5	332.1	85.3	350.3	74.4	8.7	21.6
Y325RD	3	106.6	66.9	102.0	67.5	082.4	68.1	49.7	17.7
Y345RD	7	067.7	61.7	094.9	65.8	158.9	32.6	6.9	24.7
Y346RD	3	237.1	70.6	234.6	69.5	226.8	65.0	42.5	19.1
Mean	(4)	100.1	82.3	118.0	82.1	157.2	76.1		
		K=13.4	α_{95}=26.0°	K=15.4	α_{95}=24.2°	K=5.3	α_{95}=44.0°		
South pole position				(29.2°S, 148.1°E)					
				DP=45.8°, DM=47.1°					

C. Possible primary remanence (C_3)

Core		*In situ*				In stratigraphic co-ordinates			
	n(N)	Dec. (°)	Inc. (°)	k	α_{95} (°)	Dec. (°)	Inc. (°)	k	α_{95} (°)
Y329RD	5	042.2	−24.3	10.4	24.8	059.9	−45.0	10.4	24.8
South pole position									
			(48.7°S, 035.9°E)				(35.9°S, 059.6°E)		
			dp=14.2°, dm=26.5°				dp=19.8°, dm=31.4°		

n (N): number of samples (drill cores);

α_{95}: half angle of the 95% confidence cone;

DP, DM (dp, dm): semi-major and semi-minor axis of the 95% confidence ellipse centred about the palaeopole. DP and DM are calculated from core-mean directions, and dp and dm from sample-mean.

Dec: declination

Inc: inclination

k: precision parameter

field demagnetization with a peak field of between 60 and 80 mT (600 and 800 Oe; Fig. 3a–c,e–g,i,j,l, m), although in some reddish samples thermal demagnetization, requiring only low temperature, seems to be more efficient (Fig. 3h,k). These features indicate that magnetite is likely to be the major magnetic carrier of C_1, although hematite and goethite also contributed in some of the reddish samples.

Figure 4a–f shows the sample-mean directions of the C_1 component for the six drill cores. Two important features of these directions are noted. First, these directions closely parallel the drill core axis. The mean directions from the two inclined drill cores (i.e. cores Y315RD and Y345RD) group better with other core-mean directions in drill core co-ordinates (Fig. 4g) and disperse in geographic co-ordinates (Fig. 4h). This feature strongly suggests that the remanence is a drilling-induced remanence acquired during the original diamond drilling. Second, the sample-mean directions in each drill core are remarkably well grouped ($k = 100–800$ and $\alpha_{95} < 4°$), suggesting a very effective magnetization mechanism. While such tight groupings have been observed in other sedimentary rocks, they are unusual.

Auxiliary evidence supporting the interpretation above is the difference in magnetic properties between the interior and exterior of the core samples. The natural remanent magnetization intensities of the large specimens (36 mm diameter × 32 mm height) are usually one to five times higher than those of the standard specimens drilled from the centre of the same sample. The more stable remanences (i.e. C_2 and C_3) are better defined in the standard specimens than the large ones. In other words, the intensity of the drilling-induced remanence is higher in the outer part of the cores than in the central part, which is consistent with the distribution of the induced magnetic field inside a magnetic drill barrel (McWilliams & Pinto 1988). The upward inclination of the drilling-induced remanence is consistent with the geomagnetic field being deflected by the drill barrel.

Another phenomenon consistent with the *in situ* drilling-induced remanence interpretation is that the remanence is not the result of sample preparation in the laboratory (Burmester 1977; Jackson & Van der Voo 1985). First, during the re-drilling of the 25 mm diameter cores from the centre of the original 36 mm diameter cores, each core was drilled from either the top or the bottom. If the

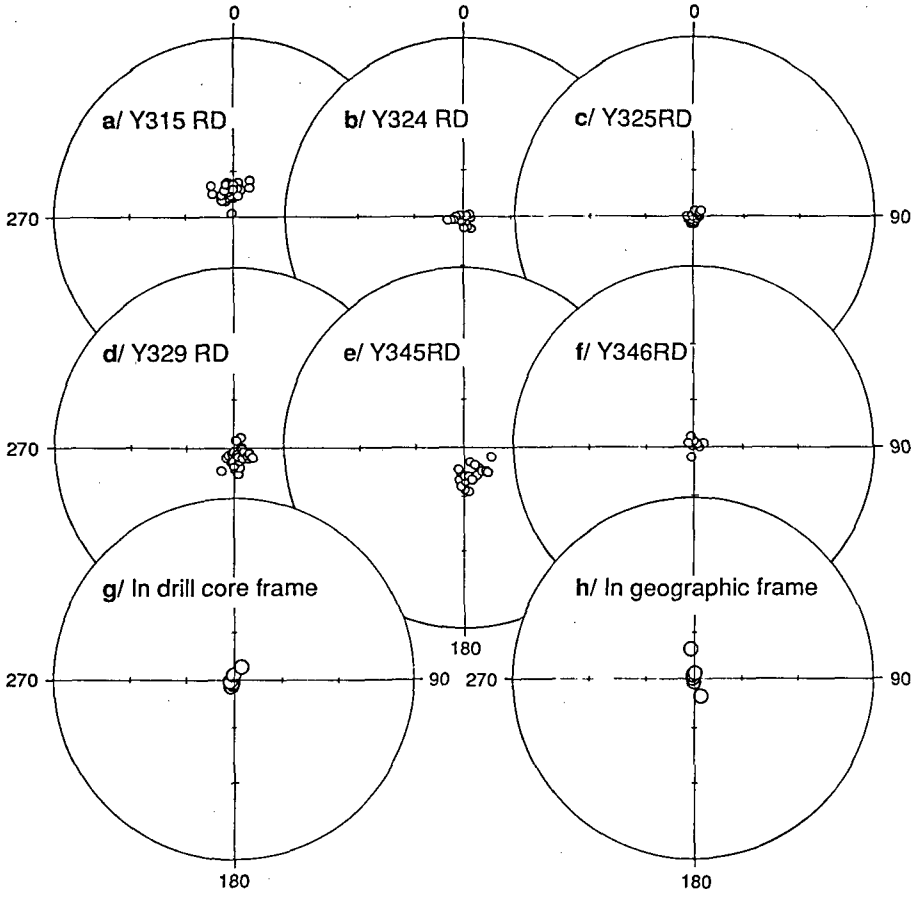


Fig. 4 Equal-angle projections of C_1. (a–f) Are the *in situ* sample-mean directions; (g) core-mean directions in core co-ordinates; (h) *in situ* (geographic) core-mean directions. Upper hemisphere projections.

measured drilling-induced remanence was formed during this process, one would expect to see opposite polarities in approximately equal numbers. Although two samples were found to possess a drilling-induced remanence of reversed polarity, this could have been the result of the drill core having been incorrectly placed in the drill tray and not having any other indication of sedimentary facing, such as cross-laminae or graded bedding. Moreover, while sawing the sub-cores into 22 mm high specimens is the final machining operation, sawing was not carried out in any consistent manner that might induce magnetizations perpendicular to the core axis. Neither does the sawing operation explain the intensity difference between the exterior and interior parts of a specimen. Therefore, machining the specimens in the laboratory can be excluded as the cause of the drilling-induced remanence.

Despite all the evidence supporting the *in situ* drilling-induced remanence interpretation, the mechanisms which cause such pervasive overprints are not well understood. Burmester (1977) and Jackson and Van der Voo (1985) reported drilling-induced remanence caused by the cutting of larger magnetite grains on the surface of samples during drilling and sawing. However, no such effect has been observed in these specimens. To test this conclusion further, three sandstone specimens were immersed in concentrated hydrochloric acid (32% HCl). These specimens are from samples in which the drilling-induced remanence accounts for at least 90% of their natural remanent magnetization. If the drilling-induced remanence occurred on the surface only, it should have been easily dissolved by acid etching (for 12–14 h, Burmester 1977). However, after a 16 h acid etching, two specimens showed a reduction in remanent intensity by 31% and one by only 4%. There was no significant change in their directions. These results indicate that the drilling-induced remanence is unlikely to be a surface feature.

A more plausible explanation for such a pervasive overprint is probably the combination of isothermal remanent magnetization and piezo-remanent magnetization, formed in an axially-symmetric strong magnetic field inside the magnetic drill barrel (McWilliams & Pinto 1988), during the initial drilling process. Similar mechanisms have been favoured recently by other workers (Kodama 1984; Audunsson & Levi 1989).

It can therefore be concluded that remanence carried by magnetite is most affected by the drilling-induced remanent magnetization, because in none of the specimens does hematite seem to have contributed much to C_1, and magnetite has not been found to carry any remanence other than C_1.

Syn-deformational overprint

C_2 was found in four out of the six drill cores, that is Y324RD, Y325RD, Y345RD and Y346RD. The proportion of samples carrying this component in the four cores is generally $<50\%$. The samples carrying C_2 are almost all of reddish colour. C_2 is most clearly identified in the re-drilled standard-size specimens rather than in the large (36 mm diameter) specimens.

C_2 was very resistant to alternating field demagnetization, but can be efficiently demagnetized by moderate to high temperature ($\sim$200–690°C, Fig. 3g–l). Therefore, hematite is likely to be the main carrier for C_2, which is consistent with the colour of the specimens.

Figure 5 shows the sample-mean directions of C_2 for the four drill cores. The four core-mean directions, with 0, 20 and 100% of bedding-correction (Fig. 6a–c), demonstate a significant dispersion using a full bedding-correction. McFadden's (1990) correlation test indicates that the four core-mean directions after bedding-correction have an apparent correlation with their bedding attitudes with more than 95% confidence, but there is no significant correlation before bedding-correction (the statistical parameter, SCOS, reaches its minimum at 2% of complete untilting). Therefore, the remanence is not likely to be of pre-folding origin. As both the Fisherian precision parameter (K) and 95% confidence circle (α_{95}) reach a maximum/minimum value after $\sim$20% of unfolding (Fig. 6d), it is concluded that the remanence was acquired just before

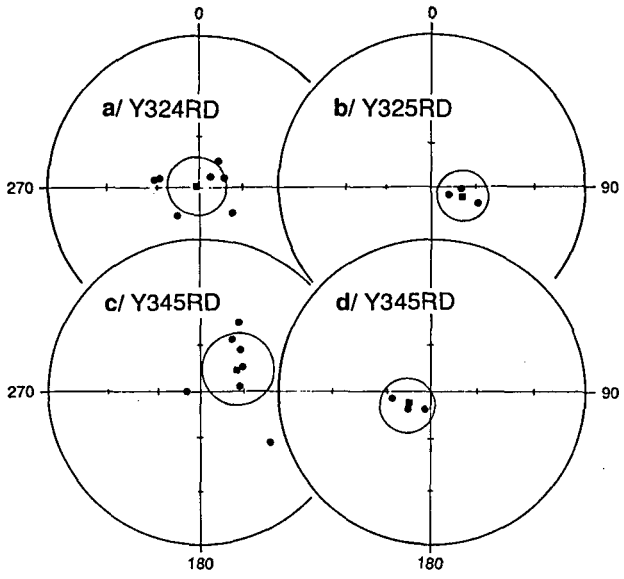


Fig. 5 Lower hemisphere equal-angle projections of the sample-mean directions of C_2. The core-means (squares) and α_{95} circles (small circles) are also shown.

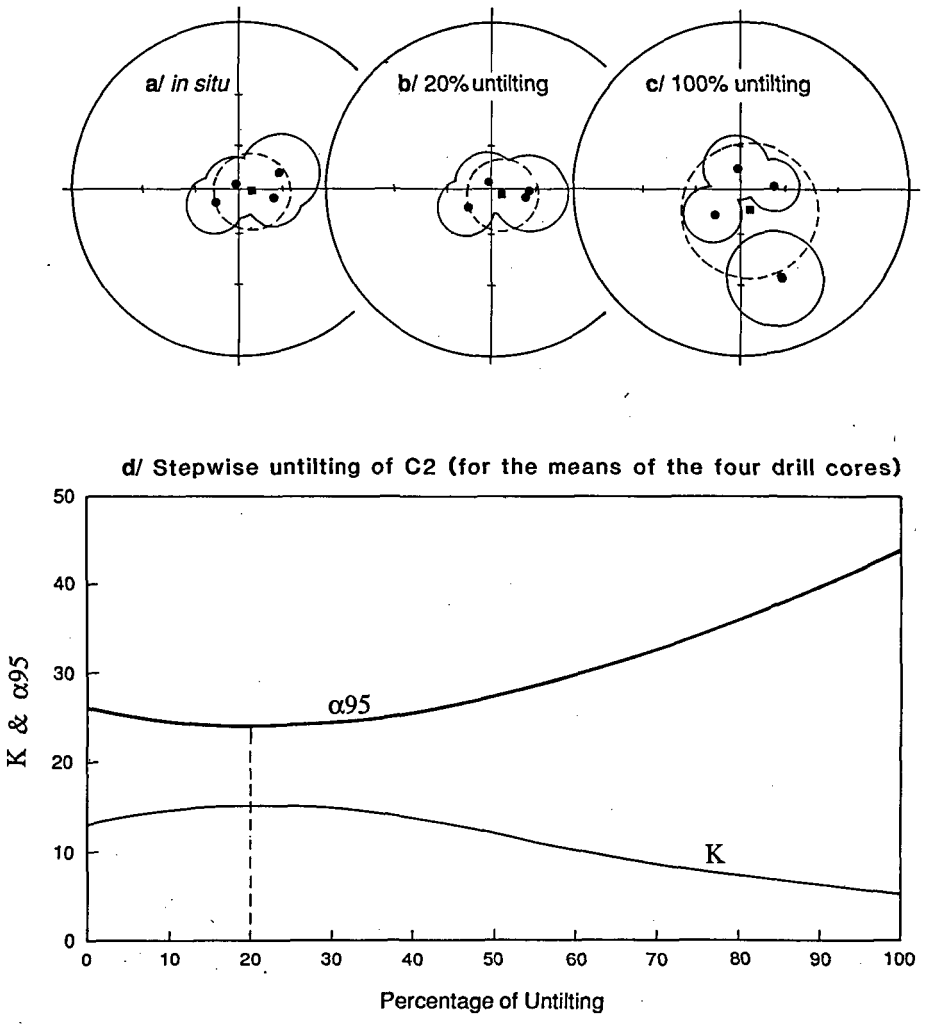


Fig. 6 (a–c) Change of directions and (d) Fisherian distribution parameters (K and α_{95}) of the C_2 core-mean directions during incremental unfolding. Dots in (a–c) are lower hemisphere equal-angle projections of the core-mean, and squares are the total mean. Small circles are projections of the α_{95} cones.

the folding finished. The syn-deformational feature, together with the direction and polarity of the new results here, indicates that the C_2 component is comparable with the mid-Carboniferous syn-deformational overprint reported from surface samples (Li *et al.* 1989).

From the Davis Gap area at the northern margin of the basin, where the Mt Eclipse Sandstone is highly folded, Li *et al.* (1989) found that the site-mean directions of the syn-deformational overprint show the best grouping after 75% of unfolding. They also reported that results from the less intensely folded localities required only 30–35% of unfolding to make the best overall population group. The latter amount of unfolding is similar to the 20% untilting required by results from this study. This suggests that rocks in the less deformed area were remagnetized during the late stage of deformation.

C_2 in this study gives a palaeomagnetic pole at 29.2°S, 148.1°E with DP = 45.8°, DM = 47.1°. This pole is plotted in Fig. 7 (pole EL-D) with Li *et al.*'s (1990) mid- to Late Palaeozoic apparent polar wander path for Australia as a reference. Pole EL-B is the preferred mid-Carboniferous overprint pole obtained from surface samples of the northern margin of the basin (Li *et al.* 1989; the other alternative pole positions, EL-A and EL-C based on two unlikely extreme structural correction models, are not shown in the figure). The EL-D pole falls on a position slightly younger than that of the EL-B pole, although the large 95% confidence oval of EL-D totally includes that of EL-B.

Possible primary remanence

In samples from core Y329RD, which are mostly red muddy siltstone, no C_2 components were found. Instead, in five samples a stable component, C_3, was found, which is directed differently from both C_1 and C_2 (Fig. 8; Table 2). The demagnetization spectrum of this component is similar to that of C_2, and hematite is its main carrier.

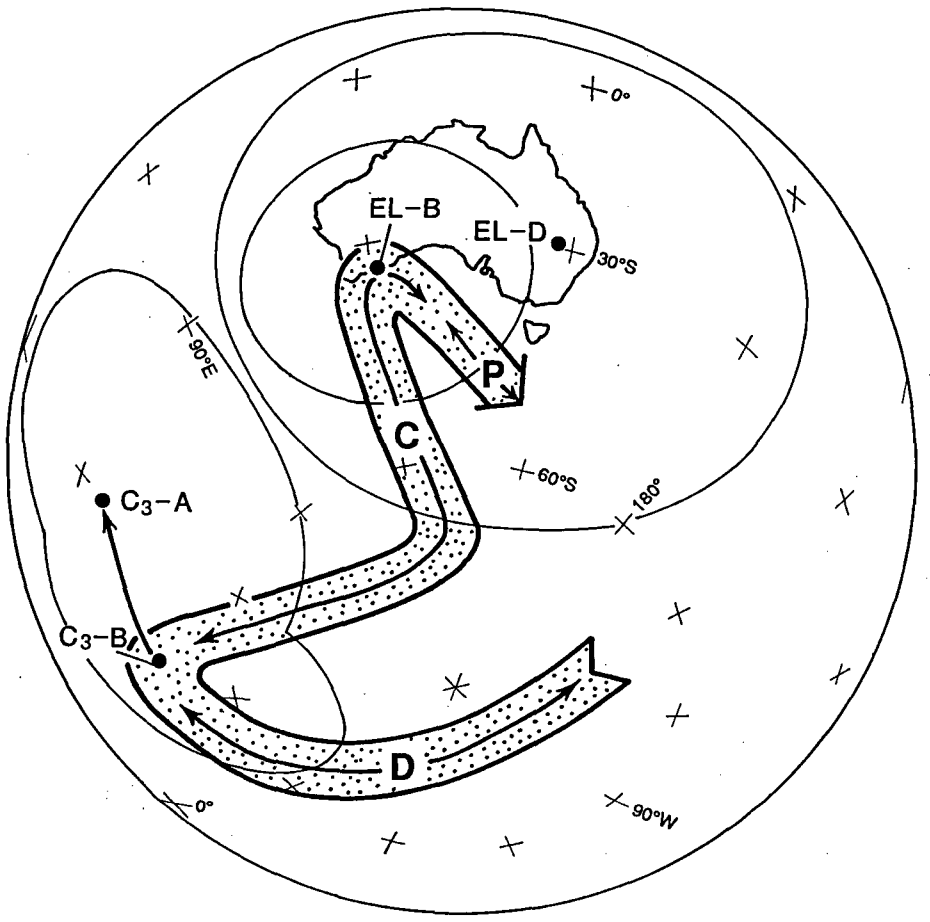


Fig. 7 Positions of poles from C_2 (EL-D) and C_3 (C_3-A and C_3-B) in the suggested Australian mid- to Late Palaeozoic apparent polar wander path (Li *et al.* 1990). C_3-B is the pole position before bedding-correction, and C_3-A is after bedding-correction. EL-B is the pole calculated from the syn-deformational remanence revealed from surface samples (Li *et al.* 1989).

Because no field test (e.g. fold test or conglomerate test, Graham 1949) is available, the palaeohorizontal and the age of this component cannot be determined with confidence. However, the difference between these directions and any Late Carboniferous or younger directions indicates that C_3 is unlikely to be a younger overprint. In fact, poles calculated from both the *in situ* directions (pole C_3-B in Fig. 7) and the bedding-corrected directions (pole C_3-A in Fig. 7) are not far away from the latest Devonian to earliest Carboniferous part of the Australian apparent polar wander path (Li *et al.* 1990). This gives encouragement that an Early Carboniferous primary palaeomagnetic pole may be obtained by further investigation of the drill cores, concentrating upon reddish materials.

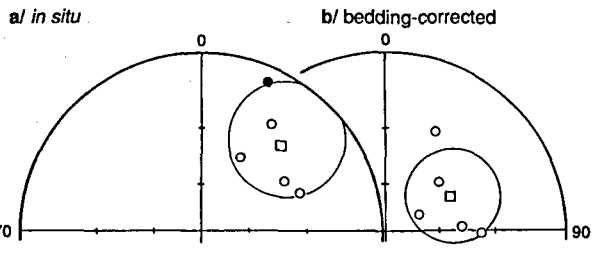


Fig. 8 Equal-angle projections of the C_3 sample-mean directions. The core-mean directions (squares) and related α_{95} circles are also shown. Open/filled symbols are projections on the upper/lower hemisphere.

DISCUSSION

Although the determination of an Early Carboniferous palaeomagnetic pole for cratonic Australia (and therefore Gondwana) has not been fulfilled, a syn-deformational overprint has been successfully isolated from a pervasive drilling-induced remanence. This overprint was acquired during the later stages of the mid- to Late Carboniferous deformation in the Ngalia Basin. As with the surface samples, hematite is the major carrier for this phase

of widespread overprint. This implies that the Alice Springs Orogeny in this region (the Mt Eclipse phase) was accompanied by a high level of oxygen in the ground water — orogenic fluids, which migrated southward from the thrusting/deforming zone at the northern margin of the basin.

The orogenic fluid migration model, as suggested by McCabe *et al.* (1989) and McCabe and Elmore (1989) for the Appalachian Basin of North America, could well be applied here. The fine-grained upper Mt Eclipse Sandstone could have functioned as a less permeable layer which has forced the orogenic fluids to migrate preferentially through the coarser grained lower part of the formation (Fig. 9). If true, this implies that the finer grained upper Mt Eclipse Sandstone has been less affected by the orogenic fluids than the lower part of the formation. The well-developed C_2 component from both the northern and southern margins of the basin (Li *et al.* 1989 and this study), as well as the lack of C_2 component and the existence of a possible primary remanence in the stratigraphically higher red muddy siltstone from core Y329RD, which was drilled closer to the centre of the basin (Fig. 1), are indeed consistent with this model. Therefore, any further work on this formation aimed at finding the primary remanence should target the sampling of fine-grained reddish material from the central part of the basin.

The fluid migration event proposed above could also have been responsible for the formation of the uranium mineralization in the lower part of the Mt Eclipse Sandstone along the northern and southern (but mainly northern) margins of the Ngalia Basin (Ivanac & Spark 1976).

The drilling-induced remanence is carried mainly by magnetite and, therefore, can usually be delineated by alternating field demagnetization. It is most likely to be a combination of isothermal remanent magnetization and piezo-remanent magnetization formed in an axially symmetric strong magnetic field, inside the magnetic diamond-drill bit, during the initial drilling process. Because the drilling-induced remanence has little impact on hematite, drill cores from hematite-bearing rocks are more likely to retain their palaeomagnetic remanences.

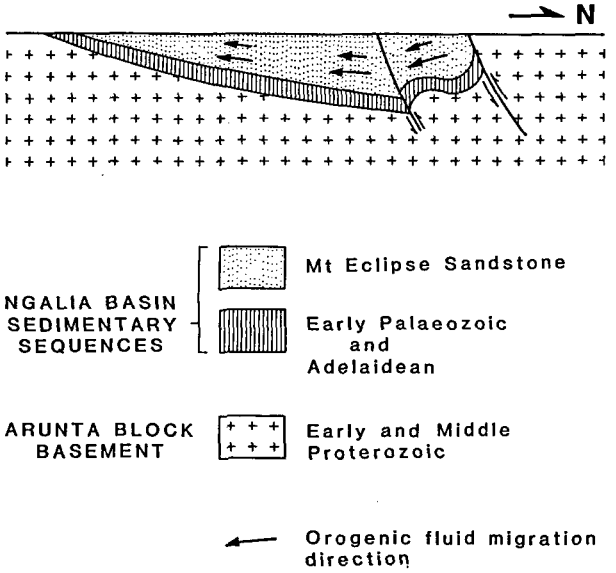


Fig. 9 Schematic model of the orogenic fluid migration path in the Ngalia Basin during the Mt Eclipse phase of the Alice Springs Orogeny. The finer grained upper Mt Eclipse Sandstone could have acted as a less permeable cap during fluid migration.

ACKNOWLEDGEMENTS

We thank Dr P. W. Schmidt and Mr D. A. Clark, both of CSIRO Division of Exploration Geoscience, for discussions during this study, and for the use of their rock magnetism laboratory. We also thank Mr A. Muraro, Agip Australia Limited, for permission to study the Ngalia Basin cores, Associate Professor B. G. Jones, University of Wollongong, for information about the orientation of the strata and the drill holes, Dr R. Thompson, Northern Territory Geological Survey, for the use of facilities in Alice Springs, and Mr M. Huddleston for assisting with sampling. Comments from Drs P. W. Schmidt, A. T. Wells and B. G. Jones improved the manuscript. The work was funded by ARC grants A38315504 and A38831488, and Macquarie University.

REFERENCES

AUDUNSSON H. & LEVI S. 1989. Drilling-induced remanent magnetization in basalt drill cores. *Geophysical Journal International* **98**, 613–622.

BURMESTER R. F. 1977. Origin and stability of drilling induced remanence. *Geophysical Journal of the Royal Astronomical Society* **48**, 1–14.

CAPUTO M. V. & CROWELL J. C. 1985. Migration of glacial centers across Gondwana during Paleozoic Era. *Geological Society of American Bulletin* **96**, 1020–1036.

COWARD D. A. 1989. *SIRO-SUS II Rock Spinner Magnetometer System User's Manual.* CSIRO report (unpubl.).

COWARD D. A., CRIPPS G. S., MAHER B. C. & RIDLEY B. H. 1985. Zero magnetic field furnace controller. *CSIRO Division of Mineral Physics 1983/1984 Biennial Report*, 26 pp.

GRAHAM J. W. 1949. The stability and significance of magnetism in sedimentary rocks. *Journal of Geophysical Research* **54**, 131–167.

IVANAC J. F. & SPARK R. F. 1976. The discovery of uranium mineralization in the Ngalia Basin, NT. *Proceedings of the Australasian Institute of Mining and Metallurgy* **257**, 29–32.

JACKSON M. & VAN DER VOO R. 1985. Drilling-induced remanence in carbonate rocks: occurrence, stability and grain-size dependence. *Geophysical Journal of the Royal Astronomical Society* **81**, 75–87.

KIRSCHVINK J. L. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. *Geophysical Journal of the Royal Astronomical Society* **62**, 699–718.

KODAMA K. P. 1984. Palaeomagnetism of granitic intrusives from the Precambrian basement under eastern Kansas: orienting drill cores using secondary magnetization components. *Geophysical Journal of the Royal Astronomical Society* **76**, 273–287.

KOPPEN W. & WEGENER A. 1924. *Die Klimate der geologischen Vorzeit.* Bornträger, Berlin.

LI Z. X., POWELL C. MCA. & SCHMIDT P. W. 1989. Syn-deformational remanent magnetization of the Mount Eclipse Sandstone, central Australia. *Geophysical Journal International* **99**, 205–222.

LI Z. X., POWELL C. MCA., THRUPP G. A. & SCHMIDT P. W. 1990. Australian Palaeozoic palaeomagnetism and tectonics — II: a revised apparent polar wander path and palaeogeography. *Journal of Structural Geology* **12**, 567–575.

MCCABE C. & ELMORE R. D. 1989. The occurrence and origin of late Paleozoic remagnetization in the sedimentary rocks of North America. *Review of Geophysics* **27**, 471–494.

MCCABE M., JACKSON M. & SAFFER B. 1989. Regional patterns of magnetite authigenesis in the Appalachian Basin: implications for the mechanism of late Paleozoic remagnetization. *Journal of Geophysical Research* **94**, 10429–10443.

MCFADDEN P. L. 1990. A new fold test for palaeomagnetic studies. *Geophysical Journal International* **103**, 163–169.

MCWILLIAMS M. & PINTO M. J. 1988. Paleomagnetic results from granitic basement rocks in the Cajon Pass scientific drill hole. *Geophysical Research Letters* **15**, 1069–1072.

SCHMIDT P. W. & EMBLETON B. J. J. 1987. A critique of paleomagnetic results from Australian Palaeozoic fold belts and displaced terranes. *In* Leitch E. C. and Scheibner E. eds. *Terrane Accretion and Orogenic Belts. Geodynamic Series* **19**, 21–30.

WELLS A. T. & MOSS F. J. 1983. The Ngalia Basin, Northern Territory: stratigraphy and structure. *Bureau of Mineral Resources Bulletin* **212**.

ZIJDERVELD J. D. A. 1967. A.C. demagnetization of rocks: analysis of results. *In* Collinson D. W., Creer K. M. and Runcorn S. K. eds *Methods in Paleomagnetism*, pp. 254–286. Elsevier Science, New York.

(*Received* 11 *September* 1990; *accepted* 6 *February* 1991)